Jay Hemnani

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Gujarat, India (Open to Relocate)

Education

Pandit Deendayal Energy University (PDEU) 2018 – 2022 | Gandhinagar, India
B.Tech in Computer Engineering | GPA: 8.7/10

- **Coursework:** Operating Systems, Database Systems, Software Development, Big Data Analytics.

Technical Skills

Languages: Python, Java, Go, JavaScript/TypeScript, C++, SQL
Backend: Node.js, Express, FastAPI, REST APIs, GraphQL, gRPC, Microservices Architecture
Distributed Systems: Apache Kafka, Redis, Message Queues, Event-Driven Architecture, Load Balancing
Databases: PostgreSQL, MySQL, MongoDB, Redis, TimescaleDB, Database Design & Indexing
Cloud & DevOps: AWS (EC2, S3, Lambda, ECS), Azure, Docker, Kubernetes, GitHub Actions, CI/CD
Tools: Git, Linux, Nginx, JWT/OAuth, API Gateway, Prometheus, Rate Limiting

Experience

Data Analyst, *Elite Hotel Group* Summer 2025 | Remote

- Defined **metrics and SLAs** with finance and operations; they became the spec for the ETL and forecasting work.
- Built **automated ETL pipelines** in SQL and Python, cutting manual data preparation by **40%** across multi-property workflows.
- Built **dashboards** in Tableau and Power BI for occupancy, revenue, and KPI tracking across the hotel properties.

Technical Consultant, *Independent* 2022 – 2024 | Remote

- Did **data analytics and pipeline consulting** for small businesses; built their reporting automation and data infrastructure.

AI/ML Intern, *Amnex* Jan – May 2022 | Gujarat, India

- Built a **fraud detection ensemble** in Python (Random Forest, XGBoost, SMOTE); **94% precision** on transaction data.
- Built **automated analytics dashboards** on statistical models for operational reporting and anomaly detection.

Software Engineering Intern, *Cactus Creatives* May – Nov 2019 | Gujarat, India

- Built a **cloud-native communication platform** for first responders: microservices on Azure with REST APIs.
- Set up **CI/CD pipelines** for automated testing and deployment, cutting deployment time by **60%**.

Projects

Kayak Travel Metasearch (team) | *Node.js, React, FastAPI, MySQL, MongoDB, Redis, Kafka, Docker*

- **3-tier system**: an API gateway handles JWT auth, rate limiting and routing to user, search and booking services; bookings use transactional inserts and inventory decrements.
- MySQL for bookings, MongoDB for analytics and logs, **Redis** for search caching. **Kafka** sits behind an interface with an in-memory stand-in for local dev.
- Python AI service with a **semantic cache** (cosine > 0.85, in Redis) that reuses LLM answers for similar queries.

Airbnb Distributed Booking (team) | *React, Node.js, MongoDB, Kafka, FastAPI, Docker, Kubernetes*

- Moved a monolith to **4 Node/Express microservices** (traveler, owner, property, booking), each with its own MongoDB, under a React/Redux frontend.
- **Kafka** booking topics with consumer groups and retry; the booking flow keeps running, degraded, if the broker is down. Docker Compose, Kubernetes manifests, JMeter load tests.
- FastAPI agent service that uses LangChain and a local Ollama model to write day-by-day travel itineraries as JSON.

CAG Deep Research | *Python, LangGraph, LangChain, Pydantic, Ollama, Groq*

- **LangGraph state machine** (7 nodes) that builds a causal graph per question; red-team and blue-team agents test each edge in parallel and a judge rules on it.
- Ports and adapters make **7 search and LLM providers** swappable by config; depth caps, per-edge limits and loop detection keep it from running forever.

Open Source Contributions

220+ merged pull requests across 37 open-source projects, including vLLM, Effect, Caddy, Lightpanda, radare2, NestJS, SeaweedFS, Jaeger and the MCP Python SDK.
A plain-English note on each one: jayhemnani.in/oss